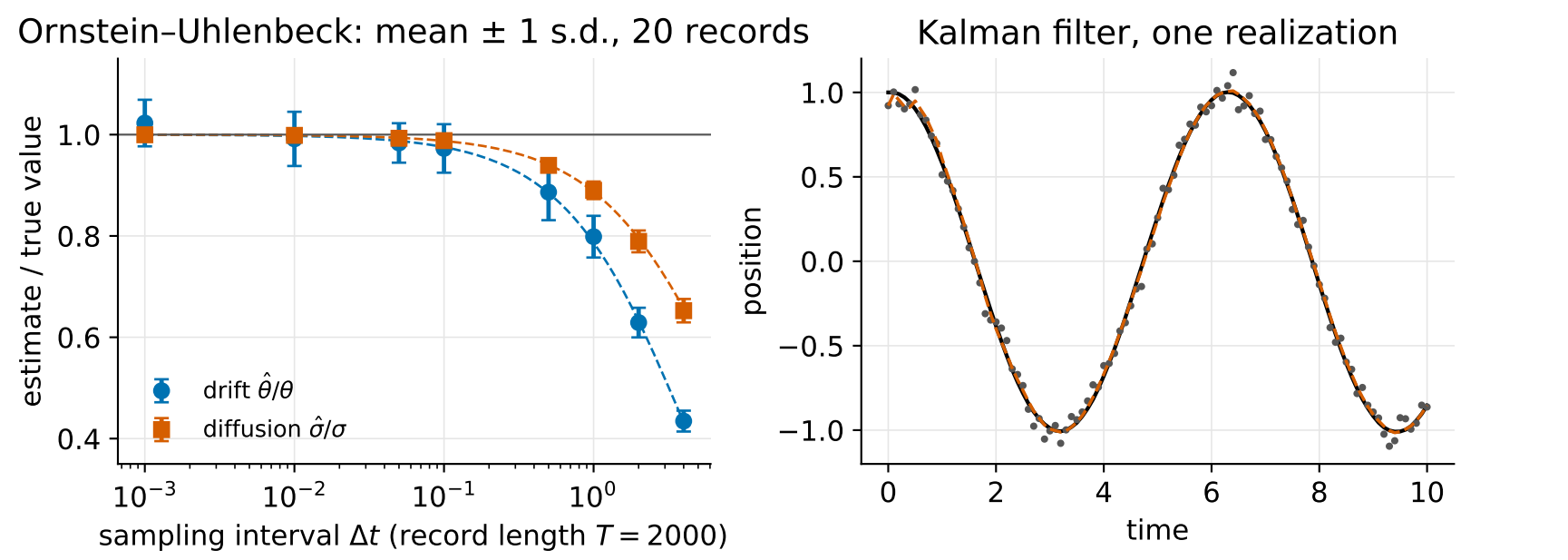
Ornstein–Uhlenbeck: mean ± 1 s.d., 20 records
estimate / true value
1.0
0.8
0.6
0.4
drift $\hat{\theta}/\theta$
diffusion $\hat{\sigma}/\sigma$
$10^{-3}$
$10^{-2}$
$10^{-1}$
$10^{0}$
sampling interval $\Delta t$ (record length $T = 2000$)
Kalman filter, one realization
position
1.0
0.5
0.0
−0.5
−1.0
0
2
4
6
8
10
time
true position
noisy observations
Kalman estimate